Strand 1: Number

Sub-strand 3: Fractions, Decimals and Percentages

Lesson 1: Fraction Arithmetic, Order of Operations, and Percentages

1. Curriculum Standard and Overview

- **Strand**: 1 — Number
- **Sub-strand**: 3 — Fractions, Decimals and Percentages
- **Content Standard**: B9.1.3.1
- **Indicators**:
- **B9.1.3.1.1** — Review fractions and solve problems involving basic operations on fractions (compare, order, convert).
- **B9.1.3.1.2** — Add, subtract, multiply and divide fractions, using the correct order of operations (BODMAS).
- **Mock-paper links**: P2 Q2(a) (mixed numbers), P2 Q4(c) (BODMAS with fractions), P1 Q12 (n/100), P1 Q26 (order fractions).

2. Preamble — What a Fraction Really Is

A fraction `(a)/(b)` answers one question: **"how many equal parts of size `(1)/(b)` do we have?"**

- **Numerator** (top): how many parts we have.
- **Denominator** (bottom): how many equal parts the whole was cut into.

So `(3)/(8)` means 3 pieces, each one eighth of a whole.

Mixed number: a whole number plus a fraction, e.g. `2(1)/(2)` (two and a half). **Improper fraction:** top ≥ bottom, e.g. `(5)/(2)`.

Convert mixed → improper: `2(1)/(2) = (2 × 2 + 1)/(2) = (5)/(2)`. *Convert improper → mixed:* `(7)/(3) = 2(1)/(3)` *(because 7 ÷ 3 = 2 remainder 1).*

3. Comparing and Ordering Fractions

3.1 Same denominator

Bigger numerator wins: `(5)/(8) > (3)/(8)`.

3.2 Different denominators — use a common denominator

To compare `(3)/(8), (1)/(2), (5)/(16)`: LCM of 8, 2, 16 is **16**.

`(3)/(8) = (6)/(16), (1)/(2) = (8)/(16), (5)/(16) = (5)/(16)`

So `(5)/(16) < (3)/(8) < (1)/(2)`. *(This is Mock P1 Q26 — option C.)*

3.3 Cross-multiplication shortcut

To compare `(a)/(b)` and `(c)/(d)`, compare `ad` and `cb`. `(3)/(8)` vs `(1)/(2)`: `3 × 2 = 6`, `1 × 8 = 8`. Since `6 < 8`, we have `(3)/(8) < (1)/(2)`.

4. The Four Operations

4.1 Addition / subtraction

Need the same denominator (use the LCM).

$\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}$

$\frac{1}{3} + \frac{1}{4} = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$

4.2 Multiplication

Multiply top × top, bottom × bottom (cancel first if you can):

$\frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$

4.3 Division

Keep, Change, Flip (multiply by the reciprocal):

$\frac{2}{5} \div \frac{14}{15} = \frac{2}{5} \times \frac{15}{14} = \frac{30}{70} = \frac{3}{7}$

4.4 Mixed numbers

Convert to improper fractions first.

$2\frac{1}{2} - \frac{1}{3} = \frac{5}{2} - \frac{1}{3} = \frac{15}{6} - \frac{2}{6} = \frac{13}{6} = 2\frac{1}{6}$

5. BODMAS with Fractions

Order of operations:

1. **B**rackets
2. **O**rders (powers, roots)
3. **D**ivision and **M**ultiplication (left to right)
4. **A**ddition and **S**ubtraction (left to right)

Mock P2 Q4(c): $\frac{2}{3} \times \frac{1}{4} + \frac{2}{5} \div \frac{14}{15}$

1. Multiplication and division **before** addition:

$\frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$ $\frac{2}{5} \div \frac{14}{15} = \frac{2}{5} \times \frac{15}{14} = \frac{30}{70} = \frac{3}{7}$

2. Add: $\frac{1}{6} + \frac{3}{7} = \frac{7}{42} + \frac{18}{42} = \frac{25}{42}$.

Answer: $\frac{25}{42}$.

6. Fractions ↔ Decimals ↔ Percentages

Fraction	Decimal	Percentage
1/2	0.5	50%
1/4	0.25	25%
3/4	0.75	75%
1/5	0.2	20%
1/8	0.125	12.5%
1/3	0.333…	33■%
3/8	0.375	37.5%

5/16	0.3125	31.25%

Convert:

- Fraction → decimal: divide top by bottom.
- Decimal → percentage: × 100.
- Percentage → fraction: ÷ 100, then simplify.

Mock P1 Q12: $(n)/(100) = 10.5 \Rightarrow n = 10.5 \times 100 = 1050$. **Answer: C.**

7. Twenty Fully Worked Examples

Example 1 — Mixed to improper

$2(1)/(2) = (5)/(2)$.

Example 2 — Improper to mixed

$(11)/(4) = 2(3)/(4)$.

Example 3 — Add same denominator

$(3)/(8) + (1)/(8) = (4)/(8) = (1)/(2)$.

Example 4 — Add different denominators

$(1)/(3) + (1)/(4) = (4+3)/(12) = (7)/(12)$.

Example 5 — Subtract fractions

$(5)/(6) - (1)/(4) = (10)/(12) - (3)/(12) = (7)/(12)$.

Example 6 — Multiply fractions

$(2)/(3) \times (1)/(4) = (2)/(12) = (1)/(6)$.

Example 7 — Divide fractions

$(2)/(5) \div (14)/(15) = (2)/(5) \times (15)/(14) = (3)/(7)$.

Example 8 — Mock P2 Q2(a)(i): Amina's water

$2(1)/(2) - (1)/(3) = (5)/(2) - (1)/(3) = (15-2)/(6) = (13)/(6) = 2(1)/(6)$ litres.

Example 9 — Mock P2 Q4(c): BODMAS

$(2)/(3) \times (1)/(4) + (2)/(5) \div (14)/(15) = (1)/(6) + (3)/(7) = (25)/(42)$.

Example 10 — Mock P1 Q26: order fractions

$(5)/(16) = (5)/(16)$; $(3)/(8) = (6)/(16)$; $(1)/(2) = (8)/(16)$. Order: $(5)/(16) < (3)/(8) < (1)/(2)$ → **C**.

Example 11 — Mock P1 Q12: n/100 = 10.5

$n = 1050$ → **C**.

Example 12 — Fraction of a quantity

$\frac{3}{8}$ of 480 = $\frac{3}{8} \times 480 = 180$.

Example 13 — Percentage of a quantity

35% of GH₵ 240 = $0.35 \times 240 = 84$.

Example 14 — Convert 0.375 to a percentage

$0.375 \times 100 = 37.5\%$.

Example 15 — Convert 12.5% to a fraction

$12.5\% = \frac{12.5}{100} = \frac{1}{8}$.

Example 16 — Fraction of a fraction

$\frac{1}{2}$ of $\frac{3}{5} = \frac{1}{2} \times \frac{3}{5} = \frac{3}{10}$.

Example 17 — Share in a ratio (preview of L5)

Share GH₵ 720 in the ratio 2:3. Total parts = 5. One part = 144. Shares: 288 and 432.

Example 18 — BODMAS with mixed ops

$\frac{1}{2} + \frac{1}{3} \times \frac{3}{4} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ (multiply first).

Example 19 — Word problem: remaining litres

A tank holds $5\frac{1}{4}$ litres. $1\frac{1}{2}$ litres leak out. How much remains? $\frac{21}{4} - \frac{3}{2} = \frac{21}{4} - \frac{6}{4} = \frac{15}{4} = 3\frac{3}{4}$ litres.

Example 20 — Word problem: class fractions

In a class of 40 pupils, $\frac{3}{8}$ are girls. How many boys? Girls = 15; boys = 25.

8. Comprehensive Exercise Sets (40 Questions)

All authentic mock items for this topic are included. Generated questions top up to 40. Spiral review ≈ 15%.

Exercise 1 — Convert fractions

1. Write $2\frac{1}{2}$ as an improper fraction.
2. Write $\frac{11}{4}$ as a mixed number.
3. Write 0.375 as a percentage.
4. Write 12.5% as a fraction in its lowest terms.
5. **Spiral review:** Write 47 300 in standard form.

Exercise 2 — Order and compare (includes mock)

1. **[Mock 2026 P1 Q26]** Order $\frac{3}{8}, \frac{1}{2}, \frac{5}{16}$ from smallest to largest.

A. $\frac{3}{8} < \frac{1}{2} < \frac{5}{16}$ B. $\frac{1}{2} < \frac{3}{8} < \frac{5}{16}$ C. $\frac{5}{16} < \frac{3}{8} < \frac{1}{2}$ D. $\frac{1}{2} < \frac{5}{16} < \frac{3}{8}$

2. Which is larger: $\frac{5}{8}$ or $\frac{3}{5}$?

3. Arrange (1)/(3), (2)/(5), (1)/(2) in ascending order.
4. **Spiral review:** Classify 0 under ■, ■, ■, ■.
5. **Spiral review:** Evaluate 5^6 ÷ 5^2.

Exercise 3 — Add and subtract fractions

1. (3)/(8) + (1)/(8)
2. (1)/(3) + (1)/(4)
3. (5)/(6) - (1)/(4)
4. 2(1)/(2) - (1)/(3) **[Mock 2026 P2 Q2(a)(i) style]**
5. **Spiral review:** Round 12.473 to 2 d.p.

Exercise 4 — Multiply and divide fractions

1. (2)/(3) × (1)/(4)
2. (2)/(5) ÷ (14)/(15) **[part of Mock 2026 P2 Q4(c)]**
3. (1)/(2) of (3)/(5)
4. 3(1)/(3) × (3)/(10)
5. **Spiral review:** Simplify (3^2)^3.

Exercise 5 — BODMAS with fractions (includes mock)

1. **[Mock 2026 P2 Q4(c)]** Evaluate (2)/(3) × (1)/(4) + (2)/(5) ÷ (14)/(15).
2. Evaluate (1)/(2) + (1)/(3) × (3)/(4).
3. Evaluate ≤ft((1)/(2) + (1)/(3)) × (6)/(5).
4. Evaluate (3)/(4) ÷ ≤ft((1)/(2) - (1)/(4)).
5. **Spiral review:** Evaluate 48 ÷ 6 + 5 × 2 - 3.

Exercise 6 — Percentages and the n/100 idea (includes mock)

1. **[Mock 2026 P1 Q12]** If (n)/(100) = 10.5, find n.

A. 0.0105 B. 0.105 C. 1050 D. 105

2. Find 35% of GH■ 240.
3. A shirt marked GH■ 80 is sold at a 15% discount. Find the sale price.
4. Write (3)/(8) as a percentage.
5. **Spiral review:** (0.07 × 0.02) ÷ 14.

Exercise 7 — Fraction of a quantity & word problems

1. (3)/(8) of 480.
2. A tank holds 5(1)/(4) litres. 1(1)/(2) litres leak out. How much remains?
3. In a class of 40 pupils, (3)/(8) are girls. How many boys?
4. Amina had 2(1)/(2) litres of water and drank (1)/(3) litres. How much is left? **[Mock 2026 P2 Q2(a)(i)]**
5. **Spiral review:** 112 flies × 0.1 kg each = ?

Exercise 8 — Mixed challenge

1. Share GH■ 720 in the ratio 2 : 3.
2. Evaluate (1)/(2) + (1)/(3) + (1)/(6) and simplify.
3. Which is larger: 45% or (9)/(20)?
4. **Spiral review:** Solve 2^(x+1) = 2^5.
5. **Spiral review:** In a market of 36 fishmongers, 28 sell mackerel, 18 sell tilapia, 13 sell both. How many sell neither?

9. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. (2 × 2 + 1)/(2) = (5)/(2).
2. 11 ÷ 4 = 2 remainder 3 → 2(3)/(4).
3. 0.375 × 100 = 37.5%.
4. (12.5)/(100) = (1)/(8).
5. 4.73 × 10^4.

Solutions to Exercise 2

1. Convert to sixteenths: 5/16 < 6/16 < 8/16 → **C**.
2. (25)/(40) vs (24)/(40) → (5)/(8) larger.
3. (1)/(3), (2)/(5), (1)/(2) (common denominator 30: 10, 12, 15).
4. ■, ■, ■.
5. 5^4 = 625.

Solutions to Exercise 3

1. (4)/(8) = (1)/(2).
2. (4+3)/(12) = (7)/(12).
3. (10-3)/(12) = (7)/(12).
4. (15-2)/(6) = (13)/(6) = 2(1)/(6) litres.
5. **12.47**.

Solutions to Exercise 4

1. (2)/(12) = (1)/(6).
2. (2)/(5) × (15)/(14) = (30)/(70) = (3)/(7).
3. (3)/(10).
4. (10)/(3) × (3)/(10) = 1.
5. 3^6 = 729.

Solutions to Exercise 5

1. (1)/(6) + (3)/(7) = (7+18)/(42) = (25)/(42).
2. (1)/(2) + (1)/(4) = (3)/(4).
3. ≤ft((5)/(6)) × (6)/(5) = 1.
4. (3)/(4) ÷ (1)/(4) = 3.
5. 8 + 10 - 3 = 15.

Solutions to Exercise 6

1. n = 10.5 × 100 = 1050 → **C**.
2. 0.35 × 240 = GH■ 84.
3. 15% of 80 = 12; price = GH■ 68.
4. (3)/(8) = 0.375 = 37.5%.
5. 0.0001.

Solutions to Exercise 7

1. (3)/(8) × 480 = 180.
2. (21)/(4) - (6)/(4) = (15)/(4) = 3(3)/(4) litres.
3. Girls = 15; boys = 25.

4. $2(1)/(6)$ litres.
5. 11.2 kg.

Solutions to Exercise 8

1. One part = 144; shares = 288 and 432.
2. $(3+2+1)/(6) = (6)/(6) = 1$.
3. $(9)/(20) = 45\% \rightarrow$ **equal**.
4. $x+1=5 \Rightarrow x=4$.
5. 3.

10. Authentic Mock Items in This Lesson

Ref	Question	Answer
P2 Q2(a)(i)	$2(1)/(2) - (1)/(3)$ litres	$2(1)/(6)$ litres
P2 Q4(c)	$(2)/(3) \times (1)/(4) + (2)/(5) \div (14)/(15)$	$(25)/(42)$
P1 Q12	$(n)/(100) = 10.5$	C — 1050
P1 Q26	Order 3/8, 1/2, 5/16	C — 5/16 < 3/8 < 1/2

11. Lesson Summary

Skill	Rule
Compare fractions	Common denominator (or cross-multiply)
Add / subtract	Same denominator first
Multiply	Top×top, bottom×bottom
Divide	Keep, Change, Flip
BODMAS	Brackets, Orders, ×÷, +–
Fraction ↔ %	×/÷ 100

Lesson prepared under `Grade9-Maths/Rules.md` *— plain English, British notation, full workings required.*